📘 **AI/ML Internship – Day 36 Notes & Tasks**

🟢 **Module 7: Train-Test Split & Model Evaluation**

📌 **Part 1: Why Model Evaluation is Important?**

👉 In Machine Learning:

- Training alone is not enough
- We must check whether model performs correctly on new data

👉 This process is called:
✅ Model Evaluation

📌 **Part 2: What is Train-Test Split?**

🔹 **Definition**

👉 Train-Test Split means dividing dataset into:

- Training Data
- Testing Data

📌 **Part 3: Why Split the Dataset?**

👉 If model sees all data during training:

- It may memorize data
- Predictions become unrealistic

👉 So:

- Train data → used for learning
- Test data → used for evaluation

📌 **Part 4: Theory Answers (IMPORTANT)**

🔹 **1. What is Train Data?**

👉 Data used to train the ML model.

🔹 **2. What is Test Data?**

👉 Data used to check model performance.

🔹 **3. What is Train-Test Split?**

👉 Process of dividing dataset into training and testing parts.

🔹 **4. Why Model Evaluation is Important?**

👉 To measure:

- Accuracy
- Performance
- Prediction quality

🔹 **5. What is Accuracy?**

👉 Accuracy shows how correct the model predictions are.

📌 **Part 5: Import Libraries**

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression
```

📌 **Part 6: Create Dataset**

```
data = {
  "Hours": [1, 2, 3, 4, 5, 6, 7, 8],
  "Marks": [20, 35, 50, 65, 80, 85, 90, 95]
}
```

df = pd.DataFrame(data)

print(df)

📌 Part 7: Features & Labels

X = df[["Hours"]]

y = df["Marks"]

👉 X = input/features
👉 y = output/labels

📌 Part 8: Splitting Dataset

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

📌 Part 9: Understanding Parameters

Parameter	Meaning
test_size=0.2	20% data for testing
random_state=42	Fixed random split

📌 Part 10: Train the Model

model = LinearRegression()

model.fit(X_train, y_train)

👉 Model learns only from training data.

📌 Part 11: Make Predictions

```
predictions = model.predict(X_test)

print(predictions)
```

📌 Part 12: Compare Actual vs Predicted

```
result = pd.DataFrame({
   "Actual": y_test,
   "Predicted": predictions
})

print(result)
```

📌 Part 13: Model Accuracy Score

```
score = model.score(X_test, y_test)

print("Accuracy:", score)
```

👉 score() gives model performance.

📌 Part 14: Complete ML Workflow

Dataset
↓
Train-Test Split
↓
Train Model
↓
Test Model
↓
Prediction
↓
Evaluation

📌 Part 15: Visualization of Train-Test Split

```
Total Dataset
|
├── Train Data (80%)
|      → learning
|
└── Test Data (20%)
       → evaluation
```

📌 Part 16: Real-World Thinking

👉 Real AI systems always use testing.

Examples:

- Face recognition
- Spam detection
- Medical prediction
- Recommendation systems

👉 Without testing:
❌ Model reliability cannot be trusted.

📌 Part 17: Overfitting Concept (Basic Introduction)

🔹 What is Overfitting?

👉 When model memorizes training data too much.

Result:

- Performs well on training data
- Performs poorly on new data

👉 Train-test split helps reduce this problem.

📌 Part 18: Why random_state is Important?

👉 Ensures:

- Same split every time
- Reproducible results

🎯 Tasks for Day 36

🧠 Task 1: Theory (Write Answers)

1. What is train data?
2. What is test data?
3. What is train-test split?
4. Why model evaluation is important?
5. What is accuracy?

💻 Task 2: Dataset Practice

1. Create student dataset
2. Create salary dataset
3. Create sales dataset

💻 Task 3: Train-Test Split Practice

1. Split dataset
2. Use test_size
3. Use random_state

💻 Task 4: Model Training

1. Train linear regression model
2. Use fit()
3. Train with training data only

💻 Task 5: Prediction Practice

1. Predict test data
2. Compare actual vs predicted
3. Print accuracy score

💻 Task 6: Result Analysis

1. Identify good predictions
2. Identify prediction errors
3. Explain accuracy meaning